Contesting an International Environmental Agreement

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Abstract

International environmental agreements (IEA) typically do not require unanimous country ratification as a condition for implementation. Rather, they set a minimum threshold number of countries required to ratify. For this strategy to make sense, there must be inherent ratification uncertainty at the country level. To study this phenomenon, we further develop the 'parallel contest' of Cole et al. (2021) to model an IEA. We allow countries to choose both the threshold number of countries required, and the target level of emissions. We derive several novel results: (1) The optimal threshold and level of emissions are chosen such that a single country is 'pivotal' in expectation; (2) The optimal threshold and level of emissions move in opposite directions i.e. "narrow and deep" vs. "wide and shallow"; and (3) expected free riding forces lead toward greater expected emission reductions and increasing the probability of an IEA passing in some form.

[Preliminary and Incomplete. Please do not circulate.]

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1 Introduction

Getting climate change under control is too difficult a task for any single country to tackle alone, above all because of the incentive for each country to free ride on the efforts of all the others. The free-rider problem arises because global environmental quality is a public good whose benefits can be enjoyed without having to pay for them, and it is this feature that creates the incentive for each country to free ride on the efforts of others. To make matters worse, the international environmental agreements (IEAs) through which countries cooperate to tackle climate change create their own problems of free-riding.[1]

As part of the response to the free-rider problem, IEAs almost always include a minimum-ratification threshold (MRT) whereby only a certain share of the countries who negotiated and signed the agreement have to actually ratify it for it to go ahead. For example, the Kyoto Protocol had 192 parties to the agreement, but only required ratification by 144 parties for it to go into force. And the Paris Climate Agreement had 195 parties but required only 55 to ratify. In fact, Rutz (2002) documents that only 2 out of the 122 IEAs that he considers had no MRT. Although the MRT represents only one dimension of the multi-faceted design of an IEA, it is now widely recognized that this is a critical element in successfully concluding an agreement (Bellelli et al. 2023).[2]

The thinking behind the MRT is that any country negotiating and signing an agreement does so in good faith that they also want to ratify it, while recognizing there is some probability that ratification may fail for some countries due to the inherent incentive to free-ride. This thinking is operationalized across the board in the game theoretic literature on the formation of IEAs by assuming that negotiating, signing, and ratifying an IEA are equivalent. We will refer to this as the 'negotiation equals ratification' assumption. This assumption has facilitated many important insights in the literature on IEAs, by making it possible to highlight the tensions that arise *between* countries in IEA formation (Black et al. 1993, Rubio and Casino 2005, Carraro et al. 2009, Courtois and Haeringer 2012).

However, the 'negotiation equals ratification' assumption creates a blind spot over critical issues that arise *within* each country created by the attempts to ratify an IEA. This blind spot in turn creates limitations in understanding how minimum-participation constraints are set in practice. The purpose of this paper is to develop a theoretical framework that enables us to relax this assumption and consider the issues that come to light for IEA formation in general, and with the MRT in particular.

The blind spot occurs under the 'negotiation equals ratification' assumption because it

[1] See Keohane et al. (1994), and especially the references and quotes cited in footnote 1.

[2] A minimum-ratification threshold is thought of more generally as the number of instruments of ratification, acceptance, approval, or accession that have to be deposited before a treaty enters into force.

implies, once an agreement has been negotiated, that the ratification probability in each and every country is equal to 1. But then, no government would put political effort or resources into supporting ratification within their own country if they knew that the MRT implied only some countries had to ratify the agreement for it to go ahead, and the ratification probability of all other countries were equal to 1. With no effort being exerted by the government, this calls into question the assumption that the ratification probability is equal to 1. Moreover, extending this logic to all countries, the likelihood that any country would ratify the IEA would be significantly reduced. This then calls into question the assumption that the ratification probability in any country is equal to 1. In summary, this logic reveals that the blind spot is over how the political economy of ratification within each country affects its probability of ratification.

In the model of IEA formation we develop, we put the probability that each country ratifies the IEA at center stage. To do this, we model IEA formation as a 'parallel contest'. In game theory, a contest is a model where two (or more) opposing interest groups (or individuals) contest a decision-maker over a decision. They do this by making financial contributions or exerting costly effort to try to move the decision probabilistically in favor of the outcome they want. The probability that either decision is taken is then determined by a contest success function. In our setting, within each country, there is a pro-IEA group that would like the IEA to be ratified, and an anti-IEA group who would like the status quo of no IEA to be maintained. Each makes financial contributions to their government to increase the probability that the government will choose the outcome they prefer: ratify the IEA or fail to ratify it, respectively. Importantly, the probability that the pro-IEA group gets the outcome that they want is also the probability that the country in question ratifies the IEA.

Although as described above the IEA that we model has features of a regular contest, the fact that there are multiple countries in the IEA means we must go further and specify it as a 'parallel contest'. In a parallel contest, there are multiple countries each attempting to ratify the IEA simultaneously. This means that the contest in each country, between each pro-IEA group and each anti-IEA group and their government over ratification, runs in each country 'in parallel'. The contests are linked by the fact that the ratification probability in each country depends on the ratification probability in all the others.

For a simple illustration of how the contests are interlinked, consider for example a pro-IEA group in one country that knows the probability of ratification in all other countries is equal to zero, and the MRT is greater than one country. Then even if the pro-IEA group is successful in getting its own government to ratify the agreement, it would be impossible for the agreement to pass into force. So there would be no gain for the pro-IEA group to

contribute in favor of ratification and they would contribute nothing. Interestingly, the same issue can arise if all other countries have a ratification probability of 1, and this time the MRT is lower than all countries. Then once again the pro-IEA group has no incentive to contribute, but this time it is because it knows the IEA will go ahead regardless of how it behaves. This illustrates in a simple way the potential complexities of how the contests across countries over IEA ratification are interlinked.

We originated the concept of a parallel contest to consider the ratification of international trade agreements (Cole et al. 2021). Based on the same logic for IEAs set out above, the ratification probability of an international trade agreement (ITA) in one country depends on the ratification probability in any other party to the agreement. However, there is a fundamental difference between IEAs and ITAs in that the latter always have to be unanimous: every country must ratify the agreement for it to pass into force. IEAs are distinctive in having a MRT that does not necessarily require ratification by all countries for it to pass into force. The logic for the unanimity requirement in ITAs seems to be that the incentive to free ride on them is minimal, because most of the benefits of an ITA are realized by members. For IEAs, by contrast, the public good nature of the global environment means that a country does not have to be a member to enjoy its benefits, creating an incentive to free ride. This raises a set of issues for analysis of IEAs that go beyond ITAs. In this paper we extend the framework of Cole et al. (2021) to consider these issues, the main one being the determination of the MRT.

Our model of IEA formation has N countries, all of which are parties to a single IEA. To introduce the role of domestic politics in IEA formation, we model the economy of each country is as in Marchiori et al. (2017). In that model, the payoff of the anti-IEA group in each country is increasing in domestic emissions, capturing in a reduced-form way that emissions tend to increase with production and profits. The payoff of the pro-IEA group in each country is decreasing in global emissions. In our model, before IEA formation takes place, each country determines its level of polluting emissions to maximize national welfare. Then, countries set up an IEA to try to reduce emissions. They jointly choose the emission level of each country that ratifies the agreement, and the MRT for the agreement to pass into force. This constitutes an agreement that all parties are prepared to sign. Each and every country then holds a contest in parallel to determine whether or not they ratify the IEA. The IEA then passes into force if enough countries ratify it to clear the MRT.

In forming the IEA, governments must agree on two elements: the agreement emission level (for each country), and the MRT. To highlight the tensions that each government faces, we assume that all countries are identical. That way, all governments want the same agreement emission level and MRT, so reaching agreement is not an issue. This set-up

enables us to focus entirely on the tensions that each government faces in setting each of the two elements.

In setting the agreement emission level, the tension that each government faces is as follows. As the agreement emission level is decreased, both the pro-IEA interest group and the anti-IEA interest group are prepared to contribute more towards their objective: the pro-IEA group because they would gain more from lower emissions, and the anti-IEA group because they would gain more from putting a stop to having to undertake them. The increase in lobbying contributions by both groups benefits the government, so lobbying creates an incentive for the government to lower the agreement emission level. In addition, the lower the agreement emission level, the greater the benefit to the government if it fails to ratify the agreement. We call this the 'expected free riding benefit'. But at the same time, a tension is created by the fact that the changes in how the lobby groups value success in their contest, which we call their 'valuations', affect the probability of ratification.

A feature of our model is that while the valuations of both groups increase as the agreement emissions level is decreased, the valuation of the anti-IEA group increases faster than that of the pro-IEA group. This reflects the idea that firms are more adversely affected by a reduction in the agreement emission level and have more resources to lobby the government against ratification than the pro-IEA group. So this leads the ratification probability to fall as the agreement emission level is reduced, and this makes the government worse off. Each government balances these tensions in setting the agreement emission level.

2 The Economy and Government Objectives

To model the economic and environmental structure of the global economy, we use the symmetric N-country model of Marchiori et al. (2017), wherein all countries are identical to one another. The government in each country $i \in N$ chooses environmental policy (EP) e_i.[3] Adopting a reduced-form approach, we assume that firms in country i will produce until they reach the emissions limit implied by e_i. Hence we equate the emissions level of country i to its EP, e_i.

Given the choice of EP by each and every country $e = (e_1, ..., e_N)$, country i receives a national benefit $B(e_i)$ that only depends on its own EP, but suffers environmental damage

[3] We use N to denote both the set and the number of countries.

$D(e)$ that depends on the EPs of all countries:

$$B_i(e) = \beta e_i - \frac{1}{2}e_i^2 \tag{1}$$

$$D_i(e) = \omega \sum_{j=1}^{N} e_j. \tag{2}$$

Higher values of parameters β and ω increase the (diminishing) marginal benefit of emissions and the (constant) marginal cost of emissions respectively.

Country i's national welfare is defined as the difference between the national benefit and damage from EP:

$$W_i(e) = B_i(e_i) - D_i(e) = \beta e_i - \frac{1}{2}e_i^2 - \omega \sum_{j=1}^{N} e_j, \tag{3}$$

where e_{-i} is the vector of environmental policies set by all countries other than country i. The emissions e_i of country i that maximize its own national national benefit $B_i(e)$, its own national welfare $W_i(e)$ and global welfare $\sum_i W_i(e)$ are $e_i = \beta$, $e_i = \beta - \gamma$ and $e_i = \beta - N\gamma$, respectively. We assume β is sufficiently large that the globally optimal emissions level for country i is strictly positive.

In each country there is a lobby group whose payoff is given by the national benefit from the emissions, and a lobby group whose payoff is given by the the damage of global emissions. Given an IEA over lowering emissions, the lobby group in country i representing the national benefit from emissions, $L_{A,i}$, will oppose the agreement and is the anti-IEA lobby group, while the lobby group representing global environmental damages, $L_{T,i}$, will support the agreement and is the pro-IEA lobby group. They do this by making contributions to government i, of $l_{A,i}$ and $l_{P,i}$ respectively.

In country i, the payoff from a global environmental policy vector e is $\pi_{A,i}(e) = B_i(e)$ for the anti-IEA lobby and $\pi_{P,i}(e) = -D_i(e)$ for the pro-IEA lobby. Both $\pi_{A,i}(e)$ and $\pi_{P,i}(e)$ are ex-post payoffs in the sense that they are determined once the policy vector e is implemented.

3 An IEA as a Parallel Contest

We now consider a proposed international agreement on emission levels. We will refer to the vector of emissions policies that exist prior to the agreement as the status-quo-emissions policy vector. We will work with a symmetrical set-up, where all countries are identical and set the same status-quo-emissions policy, e_{SQ}. Therefore, we can think of e_{SQ} as both the

emissions policy set by the representative country, and the vector of emissions policies, in the status quo. For tractability, we will assume that $e_{SQ} = \arg\max B(e) = \beta$. This simplifies the analysis a great deal without losing the main tensions of the model. We will discuss the implications of relaxing this assumption and provide numerical results in Section 6.

All N countries are seeking to form an IEA that entails an emission level vector $e_{IA} = (e_{IA,1}, ..., e_{IA,N})$, together with an MRT, $M \in \{2, ..., N\}$, whereby at least M countries must ratify the IEA for it to pass into force. In each country, an independent and *parallel* contest takes place to determine whether the IEA is ratified. Given that countries are identical, it is sufficient to choose the emission level, e_{IA}, and MRT, M, that maximizes a single representative government's expected welfare. Hence we can also think of e_{IA} as the emissions policy vector of all countries that ratify the IEA.

We can define e_{IA} as the emission level of the representative country under the IEA. It is determined as follows:

$$\begin{aligned} e_{IA} &= e_{SQ} - \gamma\omega \\ &= \beta - \gamma\omega. \end{aligned} \tag{4}$$

Thus, $\gamma\omega$ captures the amount by which e_{IA} is reduced below $e_{SQ} = \beta$. While we do not explicitly impose $e_{IA} \geq 0$, this will emerge in equilibrium because the emissions negotiated in an IEA lie between the levels that maximize national and global welfare and we have assumed that the globally optimal emission level is strictly positive.

We now model the contest over an IEA between all countries $i \in N$, drawing attention to the specific features of our parallel contest framework. Negotiation and ratification of the IEA emissions policies proceeds as follows:

Stage 1 Given $e_{SQ} = \beta$, governments announce the IEA emissions policy $e_{IA} < \beta$ and the MRT, M.

Stage 2 In each and every country $i \in N$, the anti-IEA lobby $L_{A,i}$ and the pro-IEA lobby $L_{P,i}$ simultaneously make non-negative contributions, $l_{A,i}$ and $l_{P,i}$ respectively, to their own government.

Stage 3 Each government decides whether to ratify the IEA. If M or more governments ratify it, the IEA passes into force, and the emissions policies e_{IA} are implemented in the countries that ratified the IEA. All countries that do not ratify the IEA continue to set $e_{SQ} = \beta$.

As explained above, once the lobbying contributions have been made in Stage 2, each government decides in Stage 3 whether or not to ratify the IEA. However, this decision is not

deterministic. A central feature of the contest framework is that outcomes are probabilistic. Government decision-making is influenced not only by lobbying contributions but also by an inherent element of randomness. This reflects real-world uncertainty: even when one side contributes more than the other, the outcome is not guaranteed. To model this, we adopt a standard contest success function (CSF), which in our setting maps lobby contributions into the probability of ratification. Specifically, the probability ρ_i that government i ratifies the IEA decreases with $l_{A,i}$ and increases with $l_{P,i}$.

Going beyond the standard CSF approach, and following Cole et al. (2021), we assume the government's preferences endow an 'expected head start' $\mathbb{E}h_i$ that effectively augments the contributions of *one* lobby group.[4] Given an environmental policy proposal $e_{IA} < e_{SQ}$, we say the government endows an expected pro-environment head start on $L_{P,i}$ of $\mathbb{E}h_i$ if $\mathbb{E}h_i > 0$, but an expected anti-environment head start on $L_{A,i}$ of $|\mathbb{E}h_i|$ if $\mathbb{E}h_i < 0$.

In the CSF, successful ratification of the IEA is aligned with the interest of the anti-IEA lobby:[5]

$$\rho_i\left(l_{P,i}, l_{A,i}, \mathbb{E}h_i\right) = \begin{cases} \frac{l_{P,i}+b\mathbb{E}h_i}{l_{P,i}+b\mathbb{E}h_i+l_{A,i}} & \text{if } \mathbb{E}h_i \geq 0 \\ \frac{l_{P,i}}{l_{P,i}+l_{A,i}+b|\mathbb{E}h,i|} & \text{if } \mathbb{E}h_i < 0 \end{cases}. \tag{5}$$

Here, $b \geq 0$ governs the strength of the expected head start in augmenting lobby contributions. To highlight the key tensions in the model, we will first assume that there are no head starts, i.e. $b = 0$. We now proceed by backward induction.

Stage 3. With no head starts, and given the lobbying contributions received in Stage 2, government i ratifies the IEA according to the following CSF:

$$\rho_i\left(l_{P,i}, l_{A,i}\right) = \frac{l_{P,i}}{l_{P,i} + l_{A,i}} = \frac{1}{1 + \frac{l_{A,i}}{l_{P,i}}}. \tag{6}$$

That is, government i implements the policy change from e_{SQ} to $e_{IA} < e_{SQ}$ with probability ρ_i. All countries in the set follow the same decision process.

Stage 2. Given the symmetry of our model, lobby group payoffs only depend on the number of countries and not which countries ratify the IEA. Thus, we redefine the ex-post payoff $\pi_{j,i}(e)$ for lobby group $j = A, P$ in country i as $\pi_{j,i}\left(e_i; Z\right)$, where Z is the total number of countries that ratify the IEA.

[4] Siegel (2014) introduces head starts into the all-pay auction literature. Unlike Siegel (2014), our head starts are 'expected' because policy outcomes are not known when lobby groups make contributions.

[5] Within the generalized CSF $\rho_i = \frac{s^r_{P,i}}{s^r_{P,i}+s^r_{A,i}}$, our CSF represents the 'simple Tullock contest' of $r = 1$ with 'augmented contributions' $s_{P,i} = l_{P,i} + \mathbb{E}h \cdot 1\left[\mathbb{E}h \geq 0\right]$ and $s_{A,i} = l_{A,i} + |\mathbb{E}h| \cdot 1\left[\mathbb{E}h < 0\right]$. Our augmented contributions correspond to the 'effective investments' of Rai and Sarin (2009) who axiomatize the generalized CSF and thereby build on Skaperdas (1996) who axiomatizes the CSF with $r = 1$ and $s_{g,i} = l_{g,i}$. We assume that $\rho_i > 0$ if $l_{A,i} = l_{T,i} = \mathbb{E}h_i = 0$. This nests the typical assumption that $\rho_i = \frac{1}{2}$.

Intuitively, the expected payoff $\mathbb{E}u_{j,i}$ of lobby group j in country i consists of three components. To define these, start by letting X denote the number of *other* countries (excluding i) that ratify the IEA, so that $Z = X + 1$ if country i ratifies and $Z = X$ otherwise. From the perspective of a lobby group in country i, the value of lobbying effort depends on whether country i's ratification decision is pivotal for the formation of the IEA, which in turn is governed by the distribution of X and, in particular, the probability that X takes values close to the ratification threshold M. The first component measures the lobby group's payoff when the IEA does not go ahead regardless of country i's ratification decision, which occurs when $X \leq M - 2$ other countries ratify the IEA. In this case, the ex-post lobby group payoffs are $\pi_{j,i}(e_{SQ}; 0)$. Second, the IEA goes ahead regardless of country i's ratification decision when $X \geq M$ other countries ratify the IEA. In this case, the ex-post lobby group payoffs are $\pi_{j,i}(e_{IA}; X + 1)$ if country i ratifies the IEA and $\pi_{j,i}(e_{IA}; X)$ if it does not. Third, country i's ratification decision is pivotal to whether the IEA goes ahead when $X = M - 1$ other countries ratify the IEA. In this case, the ex-post lobby group payoffs are $\pi_{j,i}(e_{IEA}; X + 1)$ if country i ratifies the IEA and $\pi_{j,i}(e_{SQ}; X)$ if it does not.

Thus, lobby group expected payoffs prior to the ratification process in each country are

$$\begin{aligned}\mathbb{E}u_{j,i} = &\sum_{r=0}^{M-2} \mathrm{pr}(X = r)\, \pi_{j,i}(e_{SQ}; r) \\ &+ \mathrm{pr}(X = M - 1)\left[(1 - \rho_i)\, \pi_{j,i}(e_{SQ}; M - 1) + \rho_i \pi_{j,i}(e_{IA}; M)\right] \\ &+ \sum_{r=M}^{N-1} \mathrm{pr}(X = r)\left[(1 - \rho_i)\, \pi_{j,i}(e_{SQ}; r) + \rho_i \pi_{j,i}(e_{IA}; r + 1)\right] - l_{j,i}. \end{aligned} \tag{7}$$

We can rewrite $L_{j,i}$'s expected payoff more compactly as consisting of three terms: (i) its expected payoff if country i does not ratify, $\mu_{j,i}$, (ii) the absolute change in its expected payoff associated with the possibility of ratification in country i, $\rho_i v_{ji}$, and (iii) its lobbying expenditure, $l_{j,i}$. That is,

$$\mathbb{E}u_{A,i} = \mu_{A,i} - \rho_i v_{A,i} - l_{A,i} \tag{8}$$

$$\mathbb{E}u_{P,i} = \mu_{P,i} + \rho_i v_{P,i} - l_{P,i}. \tag{9}$$

where

$$\mu_{ji} = \text{pr}\left(X \leq M-1\right)\pi_{ji}\left(e_{SQ},0\right) + \sum_{r=M}^{N-1} \text{pr}\left(X=r\right)\pi\left(e_{SQ},r\right) \geq 0 \tag{10}$$

$$v_{ji} = \sum_{r=M-1}^{N-1} \text{pr}\left(X=r\right)\left|\pi_{ji}\left(e_{IA},r+1\right) - \pi_{ji}\left(e_{SQ},r\right)\right| \geq 0. \tag{11}$$

The lobby group "valuations" v_{ji} are the absolute change in their expected payoffs conditional on their own country ratifying the IEA. Because an IEA will lower emissions below the status quo level, IEA ratification reduces (increases) the payoff of $L_{A,i}$ ($L_{P,i}$) and hence $\rho_i v_{j,i}$ enters its expected payoff with a negative (positive) sign.

Given the identity of which countries ratify the IEA is irrelevant, and remembering that lobby group payoffs are $\pi_{A,i} = B_i\left(e\right)$ and $\pi_{P,i} = -D_i\left(e\right)$, equations (1)-(2) imply the valuations reduce to

$$v_{A,i} = \left|\Delta B_i\left(e\right)\right| \cdot \text{pr}\left(X \geq M-1\right) \tag{12}$$

$$= \frac{1}{2}\left(\gamma\omega\right)^2 \cdot \text{pr}\left(X \geq M-1\right) \tag{13}$$

$$v_{P,i} = \left|\Delta D_i\left(e\right)\right| \cdot \left[\text{pr}\left(X \geq M-1\right) + \left(M-1\right)\text{pr}\left(X=M-1\right)\right] \tag{14}$$

$$= \gamma\omega^2 \cdot \left[\text{pr}\left(X \geq M-1\right) + \left(M-1\right)\text{pr}\left(X=M-1\right)\right]. \tag{15}$$

Because the IEA lowers each members' emissions by $\gamma\omega$, equation (13) says the value to $L_{A,i}$ of preventing this emissions reduction is increasing in $\gamma\omega$ due to the diminishing marginal benefit of emissions (i.e. the marginal benefit of emissions rises as emissions fall). It is also proportional to the probability that country i's ratification is associated with the IEA being implemented. Equation (15) says the value to $L_{P,i}$ of lower IEA emissions is proportional to the constant marginal cost of emissions ω. It is also proportional to the sum of two probability terms: (i) the probability that country i's ratification is associated with the IEA being implemented, and (ii) $M-1$ times the probability that country i's ratification decision is pivotal to whether the IEA goes ahead. The latter component reflects that country i's ratification leads to all other $M-1$ countries in the IEA cutting their emissions by $\gamma\omega$ when country i is pivotal.

Equations (8)-(9) expresses lobby group expected payoffs in a form that is standard in the contest literature, while highlighting the features newly introduced by our parallel contest framework. Following the intuition of regular contests, lobbying is costly but benefits $L_{P,i}$ ($L_{A,i}$) by increasing (decreasing) the probability of ratification in their own country and hence benefiting (suffering) from lower emissions under the IEA. However, the expected payoffs also

displays the strategic interdependencies introduced by our parallel contest framework. In particular, the expected payoffs and, hence, optimal lobbying expenditure for lobby groups in country i depend on the ratification probabilities in other countries. Intuitively, lobby groups in country i only have an incentive to lobby if their ratification decision affects the IEA. But, this is only true when at least $M-1$ other countries ratify the IEA and thus depends on the ratification probabilities in other countries. This strategic interdependency between contests across countries distinguishes our parallel contest setup from the prior contest literature. Nevertheless, standard solution techniques apply because the structure of lobby group preferences in equations (8)-(9) mirrors those of a standard contest.

Maximizing their expected payoffs, $L_{A,i}$ and $L_{P,i}$ respectively, choose lobbying contributions

$$l_{A,i} = \frac{1}{2\left(1+\frac{v_{P,i}}{v_{A,i}}\right)}\bar{v}_i \tag{16}$$

$$l_{P,i} = \frac{1}{2\left(1+\frac{v_{A,i}}{v_{P,i}}\right)}\bar{v}_i \tag{17}$$

where $\bar{v}_i = \left[\frac{1}{2}\left(\frac{1}{v_{A,i}}+\frac{1}{v_{P,i}}\right)\right]^{-1}$ denotes the harmonic mean of valuations. Intuitively, total lobbying expenditures equal half of the harmonic mean,

$$l_{A,i} + l_{P,i} = \frac{1}{2}\bar{v} = \rho v_{A,i}, \tag{18}$$

and simple algebra reveals this equals the expected harm imposed on $L_{A,i}$ by the IEA. Moreover, relative lobbying expenditures depend on relative valuations

$$\frac{l_{A,i}}{l_{P,i}} = \frac{v_{A,i}}{v_{P,i}} = \frac{1}{2}\gamma\frac{\operatorname{pr}(X \geq M-1)}{\operatorname{pr}(X \geq M-1) + (M-1)\operatorname{pr}(X = M-1)}. \tag{19}$$

Relative lobbying expenditure is increasing in γ because the marginal benefit of emissions for $L_{A,i}$ rises with larger IEA emission reductions and this reduction is proportional to γ. Relative lobbying expenditure is decreasing in $(M-1)\operatorname{pr}(X = M-1)$ because this captures the benefit to $L_{P,i}$ of country i being pivotal to IEA ratification and thus the emission reductions of all $M-1$ other IEA members.

Given lobby groups' lobbying strategies, we can now characterize the equilibrium probability of ratification.

Lemma 1. *There exists a unique interior equilibrium probability of ratification $\rho(\gamma, M)$. $\rho(\gamma, M)$ is decreasing in γ and increasing in M. This applies when M is continuous or*

discrete.

Substituting relative lobbying from equation (19) into the CSF in equation (6), the equilibrium probability of ratification is

$$\rho = \frac{2\left[1 + (M-1)\text{pr}\left(X = M-1 | X \geq M-1; \rho\right)\right]}{2\left[1 + (M-1)\text{pr}\left(X = M-1 | X \geq M-1; \rho\right)\right] + \gamma}. \tag{20}$$

Apart from the extreme case of $M - 1 = N - 1$ where the IEA requires unanimity, equation (20) only defines the equilibrium ratification probability $\rho(\gamma, M)$ implicitly. Nevertheless, Lemma 1 ensures existence of a unique interior fixed point. Moreover, $\rho(\gamma, M)$ decreases with γ because a more stringent IEA tilts relative lobbying intensity towards $L_{A,i}$ given its marginal benefit of emissions increases with emission reductions but the marginal cost of emissions for $L_{P,i}$ is constant. In contrast, $\rho(\gamma, M)$ is increasing in M because an IEA with more members tilts relative lobbying intensity towards $L_{P,i}$ given the external benefit that $L_{P,i}$ derives from the IEA depends on the magnitude of *global* emissions reduction.

Stage 1. Having fully characterized lobbying behavior and the resulting ratification probability ρ, we now turn to the government's problem. We assume the government cares about expected social welfare and lobbying contributions. For consistency with the interpretation of lobby group valuations as measuring the change in lobby group payoffs relative to the status quo emissions, we also define the governments' objective function $\mathbb{G}_i$ as its expected payoff relative to the status quo. Denoting $W_i(e_i, X)$ as social welfare from country i's emissions e_i and X other countries ratifying the IEA,

$$\begin{aligned}
\mathbb{G}_i &= \text{E}W_i + l_i - W_i(e_{SQ}, 0) \\
&= \text{pr}(X < M-1) W_i(e_{SQ}, 0) + \text{pr}(X = M-1)\left[(1-\rho) W_i(e_{SQ}, 0) + \rho W_i(e_{IA}, M)\right] \\
&+ \sum_{r=M}^{N-1} \text{pr}(X = r)\left[(1-\rho) W_i(e_{SQ}, r) + \rho W_i(e_{IA}, r+1)\right] + l_i - W_i(e_{SQ}, 0).
\end{aligned} \tag{21}$$

The right hand side of equation (21) separates country i's expected social welfare into three components. The first component represents social welfare when the IEA surely fails because less than $M - 1$ other countries ratify. The second component represents expected social welfare when country i is the pivotal country because $M-1$ other countries ratify. The third component represents expected social welfare when the IEA passes regardless of whether country i ratifies because at least M other countries ratify.

Using the definition of social welfare and the expressions for lobby group valuations, the

government's objective function $\mathbb{G}_i$ reduces to

$$\mathbb{G}_i = \rho\left(v_{P,i} - v_{A,i}\right) + \gamma\omega^2 \sum_{r=M}^{N-1} r\text{pr}\left(X = r\right) + l_i. \tag{22}$$

Relative to social welfare under the status quo emissions, the right hand side of equation (22) says the ratification process impacts the government's objective function in three ways. First, IEA ratification by country i benefits $L_{P,i}$ but hurts $L_{A,i}$. Second, regardless of whether country i ratifies, it benefits from the expected emissions reductions of other countries when at least M other countries ratify and hence the IEA surely goes ahead. Third, the government collects lobby revenue. Moreover, given equation (18) implies total lobby expenditures l_i exactly offset the expected harm $\rho v_{A,i}$ imposed on $L_{A,i}$, the government's objective function reduces to

$$\mathbb{G}_i = \rho v_{P,i} + \gamma\omega^2 \sum_{r=M}^{N-1} r\text{pr}\left(X = r\right). \tag{23}$$

Equation (23) has a striking interpretation: the government's objective function is simply proportional to expected global emissions reduction from the IEA. As just described, the second term in equation (23) is expected emission reductions by other countries when country i is not pivotal. And, using equation (15), $\rho v_{P,i}$ is the sum of expected emission cuts by country i and the expected emission cuts of all other countries when country i is pivotal. More formally, using (15), the government's objective function reduces to

$$\mathbb{G}_i = \gamma\omega^2 \left[M\rho\text{pr}\left(X = M-1\right) + \sum_{r=M}^{N-1} \left(\rho + r\right)\text{pr}\left(X = r\right) \right]. \tag{24}$$

The per country emissions cut under the IEA is $\gamma\omega$. And the square parenthesis is the expected number of countries that ratify: the first term says M countries ratify when country i is pivotal with probability $\rho\text{pr}\left(X = M-1\right)$, and the second term is the expected number of countries that ratify when i is not pivotal.[6]

Proposition **??** has a useful implication. Using the expected payoff of $\mathbb{G}_i$ given by (**??**) in the structural representation of government welfare in (**??**) we can rearrange to obtain an expression for expected free-riding, Θ_i. Thus we have the following corollary to Proposition **??**.

[6]The probability $\text{pr}\left(X = r\right)$ throughout the main text is the binomial probability of $X = r$ countries ratifying out of $N-1$ countries. See the Appendix for a proof that equation (24) is equivalent to the alternative definition of expected global emissions $\gamma\omega^2 \sum_{r=M}^{N} r\text{pr}\left(X = r\right)$ where $\text{pr}\left(X = r\right)$ is the binomial probability of $X = r$ countries ratifying out of N countries.

Corollary 1. *For any country i, given equilibrium ratification probability ρ, the expected free–riding benefit Θ_i can be quantified as*

$$\Theta_i = \gamma \sum_{r=M}^{N} r f(r; N, \rho) - \rho v_{P,i}.$$

With the analytical framework in place, the next step is to examine how the underlying valuations $v_{A,i}$ and $v_{P,i}$, and therefore the probability of ratification, respond to changes in the IEA's design parameters. Understanding these responses holds the key to characterizing strategic behaviour in our parallel contest model and the lobbying intensity in each country.

4 Normal Approximation

Due to the fact that the binomial pivotality probabilities are discrete and non-differentiable, analytic characterization of optimal design is intractable without approximation. Since our interest lies in IEAs with many potential members, the normal approximation to the binomial distribution provides an accurate and fully differentiable representation. This allows us to obtain clean comparative statics without altering the underlying economics.

To evaluate how the lobby groups' valuations, and hence the probability of ratification, respond to changes in the IEA's design parameters, we need tractable expressions for the pivotality probabilities, $\bar{F}(M-2, \rho)$ and $\tilde{f}(M-1, \rho)$, that appear in $v_{A,i}(\cdot, \rho)$ and $v_{P,i}(\cdot, \rho)$.

The main analytical hurdle we face is that the binomial distribution is discrete, and hence is not differentiable. To make analytical progress, we employ a *normal approximation* to the binomial distribution, which gives a reasonably close approximation when N is large and ρ is moderate (in particular close to 0.5). This simplification will give us closed-form expressions for $v_{A,i}(\cdot, \rho)$ and $v_{P,i}(\cdot, \rho)$ that are more transparent and allow for comparative statics.

We begin by translating the MRT into terms suitable for the normal approximation. Let $x = M - 1$ denote the number of other countries required to ratify the agreement for it to pass. Where it does not cause confusion, with some abuse of terminology, we will refer to x as the MRT. Define $\mu = \rho(N-1)$ as the expected number of ratifying countries (other than i), and $\sigma = \sqrt{(1-\rho)\mu}$ as the corresponding standard deviation of foreign ratifications under the binomial approximation. That is, σ can be understood as the dispersion in the number of other countries that ratify the IEA when each does so with symmetric probability ρ. Note that σ is measured in terms of countries. Accordingly, $\sigma = 1.2$ can be interpreted as that the foreign ratification tally typically fluctuates by just over one country around its mean, μ. We can see that σ is increasing in N, since more countries implies more scope for

fluctuation in ratification rates. And it is largest around balanced contests, whereby ρ is around 1, and shrinks as ρ goes towards 0 or 1.

We will then say that the 'standardized ratification margin', z, is

$$z(\rho) \equiv \frac{x - \mu(\rho)}{\sigma}. \tag{25}$$

This tells us how many standard deviations the margin $x = M - 1$ lies above ($z > 0$) or below ($z < 0$) the expected number of other-country ratifications μ. We will refer to z as the 'standardized gap' for short. If $z > 0$ then the expected number of foreign ratifications falls short of $M - 1$. When $|z|$ is close to zero, a single country's ratification has the largest marginal effect on the probability that the IEA enters into force. When $z \gg 0$ or $z \ll 0$ the marginal effect is small: in the first case, the other countries are likely to be too few relative to x; in the second they are more than enough relative to x.

Next, we link the normal distribution to pivotality probabilities. Under the normal approximation, we write

$$\bar{G}(z) = 1 - \Phi(z)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function. The term $\bar{G}(z)$ gives the foreign tail probability, that is, the probability that at least $x = M - 1$ of the other $N - 1$ countries ratify the IEA. It represents the upper tail of the normal approximation of the normal distribution of foreign ratifications (by countries other than i). The overall probability that the IEA passes is therefore $\rho\bar{G}(z)$, since country i must also ratify the IEA. When $z = 0$, the foreign tail probability is $\bar{G}(0) = \frac{1}{2}$, implying an overall pass probability of $\rho/2$. If z is large and positive (negative), then the foreign tail probability, and hence the overall pass probability is well below (above) 50%.

The function $\bar{G}(z)$ therefore plays the role of $\bar{F}(M - 2, \rho)$ in the exact binomial model, giving the probability that the MRT M is met or exceeded by the foreign countries. Multiplying by ρ then gives the the overall probability that the IEA passes.

In addition,

$$g(z) = \frac{\phi(z)}{\sigma}$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ is the standard normal probability density function (PDF). The quantity $g(z)$ therefore gives the approximate density, scaled by σ, of having exactly $M - 1$ foreign ratifications under the normal approximation. The corresponding hazard ratio, or Mills ratio, is $\tilde{g}(z) = \phi(z)/\bar{G}(z)$, which measures the slope of the tail probability relative to its level. This plays the role of $\tilde{f}(M - 1, \rho)$ in the exact binomial model, representing the (scaled) probability mass at exactly $M - 1$ foreign ratifications.

Using these definitions, the valuations of the lobby groups simplify to

$$v_{A,i} = \frac{\omega^2\gamma^2}{2}\bar{G}(z), \qquad v_{P,i} = \omega^2\gamma\left[1 + x\tilde{g}(z)\right]\bar{G}(z).$$

These formulas make the economics of pivotality explicit. Both valuations rise with $\bar{G}(z)$, the probability that country i's ratification decision is pivotal to the IEA passing. In $v_{A,i}$ the term in γ is squared, reflecting how damages from stricter emissions cuts grow faster for the firms that $L_{A,i}$ represents. In $v_{P,i}$, the term in γ is linear, but gains an extra term $x\tilde{g}(z)$, because members of $L_{P,i}$ value not just domestic emission cuts but also those in the other x members if the IEA passes.

The equilibrium probability of ratification is determined endogenously by the relative strengths of the two lobby groups. It takes the form:

$$\rho = \frac{R\left(x, z(\rho)\right)}{R\left(x, z(\rho)\right) + \gamma} \tag{26}$$

where $R\left(x, z(\rho)\right) \equiv 2\left(1 + x\tilde{g}(z\left(\rho\right))\right)$. This expression captures the key trade-off in the model. Increasing emission stringency under the IEA (higher γ) reduces ρ by intensifying anti-IEA lobbying more than pro-IEA lobbying. Conversely, increasing the MRT x makes ratification mechanically harder, but also increases the benefit to the pro-IEA group if the agreement passes, thus increasing ρ.

Equation (26) characterizes the equilibrium probability of ratification as a fixed point, since the standardized gap z depends on ρ through the expected number of foreign ratifications. Holding this feedback fixed, an increase in emissions stringency γ lowers ρ by strengthening anti-IEA lobbying relative to pro-IEA lobbying. Conversely, for a given ρ, an increase in the MRT x raises ρ by increasing the valuation of ratification for the pro-IEA group through the pivotality term $x\tilde{g}(z)$. As in the binomial case analyzed above, the equilibrium condition (26) defines a decreasing fixed-point mapping in ρ under the normal approximation, so the comparative-static effects described above carry through once the dependence of z on ρ is taken into account. This yields the following result.

Proposition 2. *The probability of ratification ρ is decreasing in the IEA emission stringency parameter γ and increasing in the MRT x.*

This result reflects two opposing forces. On the one hand, an IEA that requires greater emissions reductions elicits stronger lobbying from both sides, but the anti-IEA group responds more aggressively, lowering the probability of ratification. On the other hand, a higher MRT increases the strategic value of participation for the pro-IEA group, since the probability of the IEA being implemented becomes more sensitive to marginal ratifications.

In summary, the analysis so far shows how strategic interdependencies across countries emerge through lobbying incentives, shaping the design parameters for an IEA. These insights serve as a bridge to each government's optimization problem, which we turn to in the next section.

5 Comparative Statics and IEA Design Implications

Now apply the normal approximation to the government's objective function. We substitute $\bar{F}(x-1) \approx \bar{G}(z)$ into (**??**) to obtain:

$$\mathbb{G} = \gamma \rho N \bar{G}(z) \tag{27}$$

To interpret this expression, begin with ρN. This captures the direct benefit to country i when the IEA passes and it is a participant. Since ρ is the probability that country i ratifies, and N is the total number of countries, ρN gives the expected number of countries participating when country i is part of the IEA. Multiplying by $\gamma \bar{G}(z)$ gives the expected welfare benefit from global emissions reductions when country i contributes to the formation of the IEA. This captures the effect on each government's welfare of IEA formation in our framework.

We now examine how changes in the design parameters of the IEA, the MRT x and the emission stringency γ, affect the equilibrium outcome. To do so, we derive the first-order conditions (FOCs) that characterize each government's best response with respect to these design parameters.

The FOCs are

$$\frac{d\mathbb{G}}{d\gamma} = \frac{\partial \mathbb{G}}{\partial \gamma} + \frac{\partial \mathbb{G}}{\partial \rho}\frac{\partial \rho}{\partial \gamma}$$
$$\frac{d\mathbb{G}}{dx} = \frac{\partial \mathbb{G}}{\partial x} + \frac{\partial \mathbb{G}}{\partial \rho}\frac{\partial \rho}{\partial x}$$

Evaluating the FOCs at an interior optimum and multiplying the first FOC through by $\partial\rho/\partial\gamma$ and the second FOC through by $\partial\rho/\partial x$, yields the equilibrium condition

$$\frac{\partial \mathbb{G}}{\partial \gamma}\frac{\partial \rho}{\partial x} = \frac{\partial \mathbb{G}}{\partial x}\frac{\partial \rho}{\partial \gamma}$$

It is more intuitively appealing to express this condition in terms of elasticities. To do so, first define $\varepsilon_{\gamma}^{\mathbb{G}} \equiv \partial\mathbb{G},/\partial\gamma \cdot \gamma/\mathbb{G}$ as the elasticity of government welfare with respect to emissions stringency. Next, $\varepsilon_{x}^{\mathbb{G}} \equiv \partial\mathbb{G},/\partial x \cdot x/\mathbb{G}$ is the elasticity of government welfare with respect

to the MRT. Multiplying both sides through by $(\gamma/\mathbb{G})$ we can write the above equilibrium condition as

$$\varepsilon_{\gamma}^{\mathbb{G}}\frac{\partial\rho}{\partial x}=\frac{\gamma}{x}\varepsilon_{x}^{\mathbb{G}}\frac{\partial\rho}{\partial\gamma}$$

The equilibrium condition can be interpreted as a balance of elasticities. Accordingly, we will refer to it as the 'equilibrium balance condition'. The left-hand side captures how sensitive government welfare is to a marginal change in stringency, γ, scaled by the induced change in the ratification probability when the MRT, x, changes. The right-hand side captures the converse channel: the government-welfare sensitivity to a marginal change in the MRT, scaled by the effect of changing stringency on ratification. The condition therefore requires that, in equilibrium, the marginal effect of stringency and the marginal effect of participation are the same.

Now let us use the model to evaluate the elasticities, taking $\varepsilon_{\gamma}^{\mathbb{G}}$ first. In our set-up, $\mathbb{G}$ is linear in γ, holding ρ fixed. Therefore, $\varepsilon_{\gamma}^{\mathbb{G}}=1$. To compute $\varepsilon_{x}^{\mathbb{G}}$, using (27), we differentiate $\mathbb{G}$ with respect to x to obtain

$$\varepsilon_{x}^{\mathbb{G}}=\left[\gamma\rho N\bar{G}_{z}\frac{dz}{dx}\right]\frac{x}{\gamma\rho N\bar{G}\left(z\right)}=-\frac{x}{\sigma}\tilde{g}(z),$$

where $\bar{G}_{z}=\frac{d\bar{G}(z)}{dz}=-\phi\left(z\right)$, and the second equality follows because $dz/dx=1/\sigma$, $\bar{G}'\left(z\right)dz/dx=-\phi\left(z\right)/\sigma=-g\left(z\right)$.[7]

Using our solutions for $\varepsilon_{\gamma}^{\mathbb{G}}$, and $\varepsilon_{x}^{\mathbb{G}}$ in the above equilibrium balance condition, we can solve explicitly for the equilibrium value of z, as follows. Begin by rewriting the equilibrium balance condition using our solutions for $\varepsilon_{\gamma}^{\mathbb{G}}$ and $\varepsilon_{x}^{\mathbb{G}}$:

$$\frac{\partial\rho}{\partial x}=-\frac{\gamma}{\sigma}\tilde{g}(z)\frac{\partial\rho}{\partial\gamma}.$$

For the purpose of deriving the equilibrium balance condition, it is sufficient to work with the partial derivatives of (26) holding z fixed, since the dependence of z on ρ enters symmetrically in both $d\rho/dx$ and $d\rho/d\gamma$ and therefore cancels in the ratio of these derivatives.

Based on (25) and (26), we have

$$\frac{\partial\rho}{\partial x}=\frac{\gamma R_{x}}{\left(R\left(x,z\right)+\gamma\right)^{2}},\ \frac{\partial\rho}{\partial\gamma}=-\frac{R\left(x,z\right)}{\left(R\left(x,z\right)+\gamma\right)^{2}},$$

where $R_{x}=2\left(\tilde{g}(z)+x\tilde{g}_{z}\frac{\partial z}{\partial x}\right)=2\left(\tilde{g}(z)+\frac{x}{\sigma}\tilde{g}_{z}\right)$ because $\partial z/\partial x=1/\sigma$. Plugging this into the

[7] Throughout the analysis, we will adopt the convention that a single letter subscript on a function denotes the first derivative of that function with respect to the variable denoted by that letter.

equilibrium balance condition and cancelling common factors yields

$$\frac{\tilde{g}(z) + \frac{x}{\sigma}\tilde{g}_z}{1 + x\tilde{g}(z)} = \frac{1}{\sigma}\tilde{g}\left(z\right).$$

Using the Mills-ratio identity $\tilde{g}_z = -z\tilde{g}(z) + (\tilde{g}(z))^2$, this simplifies to the necessary condition for equilibrium,

$$z = \frac{x - \mu\left(\rho\right)}{\sigma} = \frac{\sigma - 1}{x}. \tag{28}$$

This condition equates two forces. Raising the MRT x makes passage less likely (since $\partial z/\partial x = 1/\sigma$, holding ρ fixed), but it also raises each country's pivotality in its domestic contest, which increases the ratification probability ρ. At the optimum these effects exactly offset, yielding

$$xz = \sigma - 1,$$

i.e. the standardized gap (in s.d. units) $\times$ margin (in countries) equals dispersion (in countries) minus one country. This equilibrium condition says the MRT must be placed at the point where the marginal effect of adding an additional required ratifier exactly balances the marginal reduction in the probability that the IEA enters into force. Intuitively, the MRT is chosen so that a single country is pivotal in expectation.

Equivalently,

$$x - \mu = \frac{\sigma(\sigma - 1)}{x},$$

so the required MRT lies above the expected number of foreign ratifications when the dispersion in foreign ratifications exceeds one country ($\sigma > 1$), below it when the dispersion is less than one country ($\sigma < 1$), and exactly at it when $\sigma = 1$.

Taking γ as given, each equilibrium pair (x, ρ) satisfies both (26) and (28). As γ increases, ρ falls, hence $\mu(\rho) = (N-1)\rho$ falls. The effect on $\sigma(\rho) = \sqrt{\rho(1-\rho)(N-1)}$ depends on where ρ sits relative to $\frac{1}{2}$:

$$\frac{d\sigma}{d\gamma} \gtrless 0 \text{ iff } \rho \lessgtr \tfrac{1}{2}, \quad \frac{d\sigma}{d\gamma} = 0 \text{ at } \rho = \tfrac{1}{2}.$$

Since $xz = \sigma(\rho) - 1$, a fall in σ shifts the right-hand side down and, together with a resultant fall in μ, induces a lower margin x (hence a lower MRT $M = 1 + x$). Thus, for $\rho < \frac{1}{2}$ we have $dx/d\gamma < 0$. When $\rho > \frac{1}{2}$, the σ term works in the opposite direction while it is still the case that the fall in μ pushes x down; the net effect on x is ambiguous in sign and depends on parameters. It is small when ρ is close to $\frac{1}{2}$, where σ changes little. In all cases, the equilibrium balance condition provides the locus of (x, ρ) pairs that trade off the ratification

probability against pivotality.

To determine the equilibrium value of γ, we return to the FOC above, which we now solve explicitly as

$$\frac{d\mathbb{G}}{d\gamma} = \rho N \bar{G}(z) + \gamma N \left(\bar{G}(z) - \rho\phi(z) z_\rho\right) \cdot \left(-\frac{R(x,z)}{(R(x,z)+\gamma)^2}\right) = 0$$

Rearranging,

$$\frac{\rho\bar{G}(z)}{\bar{G}(z) - \rho\phi(z) z_\rho} = \frac{\gamma R(x,z)}{(R(x,z)+\gamma)^2} \tag{29}$$

where

$$z_\rho = -\frac{N-1}{\sigma} - z\frac{(N-1)(1-2\rho)}{2\sigma^2}$$

The left hand side of (29) is the cost-benefit ratio of increasing γ inside the government payoff term $\rho\bar{G}(z)$, taking account of how a change in ρ changes z. The right hand side captures how a marginal increase in γ depresses ρ via the contest success function (26). An interior optimum in γ equates these.

We can now characterize the design of an optimal IEA in terms of equilibrium, as follows.

Proposition 3. *For fixed N, at any interior optimum for (γ, x) (with $x \equiv M-1$):*

1. *The MRT, x, satisfies*

$$z = \frac{x-\mu(\rho)}{\sigma(\rho)} = \frac{\sigma(\rho)-1}{x}, \text{ or equivalently, } xz = \sigma(\rho) - 1.$$

   Hence $x \gtrless \mu(\rho)$ iff $\sigma(\rho) \gtrless 1$; at $\sigma(\rho) = 1$ we have $x = \mu(\rho)$.

2. *Along the equilibrium locus defined by the CSF $\rho = R(x,z)/(R(x,z)+\gamma)$, the stringency γ is determined by (29). Moreover:*

   (i) if the equilibrium satisfies $\rho \leq \frac{1}{2}$, then the continuous optimum x decreases with γ, so the optimal MRT $M = 1 + x$ is nonincreasing in γ;

   (ii) if the equilibrium satisfies $\rho > \frac{1}{2}$, the effect of γ on x (and thus on M) is ambiguous, depending on parameters.

This result says that the government's payoff, as given by (27), implies a simple equilibrium relationship: the optimal MRT x and standardized gap z satisfy $xz = \sigma(\rho) - 1$. This pins down where the MRT sits relative to the expected number of other-country ratifications: $x \gtrless \mu(\rho)$ when the dispersion of foreign ratifications (in countries) satisfies $\sigma(\rho) \gtrless 1$.

Although stringency γ does not appear in this ratio, it shifts equilibrium via the CSF: when γ is increased, ρ decreases, which moves (μ, σ). For $\rho \leq \frac{1}{2}$, both μ and σ decrease with an increase in γ. Along $xz = \sigma(\rho) - 1$, this implies that x decreases in γ. For $\rho > \frac{1}{2}$, μ decreases while σ increases. The two channels work in opposite directions, so the net effect on x is ambiguous and depends on parameters.

While the equilibrium relationship $xz = \sigma(\rho) - 1$ and the CSF pin down (x, ρ) jointly, closed-form expressions for ρ (and hence for (μ, σ, z) and the optimal MRT $M = 1 + x$) require us to fix N. Throughout our analysis, we will work in the normal regime: N is large enough that the binomial tail is well approximated by its normal counterpart. This allows us to write $\bar{F}(x-1) \approx \bar{G}(z)$ with $z = (x - \mu(\rho))/\sigma(\rho)$ and to do calculus on $\mathbb{G} = \gamma\rho N\bar{G}(z)$.

5.1 Complete Solution for large N

To see the economic structure of the equilibrium more transparently, we now analyze the large-N case, which cleans away inessential finite-sample features and reveals the underlying forces driving IEA design.

In the range where N is large, the equilibrium relationship $xz = \sigma(\rho) - 1$ and the CSF determine (x, ρ) jointly. It will also be helpful in characterizating this solution to use the fact that $z \to 0$ as N becomes large.

To see this, first note that the mean number of ratifications $\mu(\rho) = \rho(N-1)$, scales with N, while the dispersion $\sigma(\rho) = \sqrt{\rho(1-\rho)(N-1)}$ scales with the square root of N. Moreover, like μ, the MRT x also necessarily grows in proportion to N. As a result, the equilibrium solution for the standardized margin given by Proposition 3(i), $z = (\sigma(\rho) - 1)/x$, becomes small when N is large: the numerator rises with the square root of N, whereas the denominator rises linearly with N. This makes the difference between x and $\mu(\rho)$ negligible, so that in the normal-approximation regime (large but finite N), we can treat the two as effectively equal. In this sense, the equilibrium condition places the MRT almost exactly at the expected number of foreign ratifications, i.e. $x \simeq \mu(\rho)$.

We can also establish that $z \to 0$ from above. Since $\sigma(\rho) \to \infty$ as N grows (for interior $\rho \in (0, 1)$), we have that $\sigma(\rho) - 1 > 0$ for all sufficiently large N. And because $x \geq 0$, the equilibrium relationship $z = (\sigma(\rho) - 1)/x$ forces $z > 0$ for large N. We can also conclude from this that $x \to \mu(\rho)$ from above. Henceforth, we will simplify the analysis by assuming that $x = \mu(\rho)$ and $z = 0$.

When $z = 0$, we are at exactly the mean of the standard normal distribution. So $\Phi(0) = 0.5$, and this implies that the foreign tail probability, $\bar{G}(0) = 1 - \Phi(0) = 0.5$, and also that $\phi(0) = 1/\sqrt{2\pi} \simeq 0.3989$. Plugging these into the Mills ratio, $\tilde{g}(0) = \phi(0)/\bar{G}(0) = \sqrt{2/\pi} \simeq$

0.798. Intuitively speaking, as z tends to 0 in equilibrium, the foreign tail probability, $\bar{G}(z)$, tends to 1/2. Hence the overall probability that the IEA passes is $\rho\bar{G}(z) \simeq \rho/2$. The slope of the overall pass probability with respect to z is $-\rho\phi(z)$, given by the hazard function $\tilde{g}(0) = \sqrt{2/\pi}$ enters the elasticities via $\bar{G}'(z) = -\phi(z)$ and $\tilde{g}(z) = \phi(z)/\bar{G}(z)$.

Substituting $z = 0$ and $x = \mu(\rho)$ into the contest success function,

$$\rho = \frac{R(x,z)}{R(x,z)+\gamma} = \frac{2(1+x\,\tilde{g}(0))}{2(1+x\,\tilde{g}(0))+\gamma},$$

and recalling that $x = \rho(N-1)$, we obtain the fixed-point relation

$$\rho = \frac{2\big(1+\rho(N-1)\tilde{g}(0)\big)}{2\big(1+\rho(N-1)\tilde{g}(0)\big)+\gamma}.$$

Solving for ρ yields

$$\rho = \frac{1}{2}\left(1 - \frac{\sqrt{2\pi\sigma^2}}{N-1}\right),$$

and hence

$$x = \tfrac{1}{2}\left[(N-1) - \sqrt{2\pi\sigma^2}\right], \qquad \gamma = \sqrt{\tfrac{2}{\pi}}\frac{\sigma}{\rho^2}.$$

where $\sigma(\rho) = \sqrt{\rho(1-\rho)(N-1)}$. These solutions describe an equilibrium outcome in which each country is pivotal in expectation, the IEA passes with roughly half the potential members ratifying ($\rho \to 1/2$ from below), and stringency γ rises smoothly with N through its dependence on σ. We summarize our analysis as follows.

Proposition 4. *The equilibrium values of ρ, x and γ can be characterized by*

$$\rho = \frac{1}{2}\left[1 - \frac{\sqrt{2\pi\sigma^2}}{(N-1)}\right] \to \frac{1}{2} \text{ as } N \text{ becomes large.} \tag{30}$$

$$x = \frac{1}{2}\left[(N-1) - \sqrt{2\pi\sigma^2}\right] \tag{31}$$

$$\gamma = \sqrt{\frac{2}{\pi}}\left[\frac{\sigma}{\rho^2}\right] \tag{32}$$

The equilibrium solutions place the IEA in a balanced position, with the probability of ratification close to one half. This outcome arises from the structure of the government's payoff function, (27), which depends on the expected global abatement benefit from the IEA.

The government's choice of (γ, x) reflects a trade-off between the value generated by successful cooperation and the probability that the agreement achieves the MRT. The equilibrium condition $xz = \sigma(\rho) - 1$ captures this trade-off: increasing x raises the required level

of participation but lowers the probability of entry into force, while increasing γ raises the value of the agreement but also reduces the likelihood that it is ratified. The equilibrium arises at a point where the marginal welfare gains from increasing either γ or x are exactly offset by the corresponding marginal reduction in the probability that the IEA enters into force.

Because the equilibrium is interior, the implied probability of ratification ρ lies close to one half, but this is an outcome of expected-welfare maximization rather than a policy target. An increase in stringency γ lowers ρ through the contest success function, thereby reducing both the mean and the dispersion of expected foreign ratifications (μ, σ). Under the equilibrium condition $xz = \sigma(\rho) - 1$, these changes require a lower MRT x to restore the balance between the expected benefit of greater stringency and the reduced likelihood that the IEA enters into force. Hence, in the welfare-maximizing equilibrium, x or γ act as complementary design instruments, moving in opposite directions whenever $\rho \leq \frac{1}{2}$.

The large-N solution shows that equilibrium settles at a point where the expected and required ratifications almost coincide ($x \simeq \mu$). The government's optimization equates the marginal abatement benefit from increasing γ with the marginal reduction in the probability that the IEA enters into force. In this configuration, $\rho N \bar{G}(z)$ measures the expected number of participants in the IEA, while γ reflects the value of each participant's emissions reduction. The resulting equilibrium delivers a ratification probability close to one half because that is where two economic forces balance endogenously: increasing γ raises the benefit from successful cooperation but lowers the probability of entry into force, while increasing x raises the required level of participation but also makes the IEA harder to achieve. The equilibrium value of $\rho \simeq \frac{1}{2}$ therefore reflects an interior balance between expected benefit and likelihood of entry into force within each government's expected-welfare objective.

Since the equilibrium probability of ratification lies close to $\rho = \frac{1}{2}$ in the large-N case, it is informative to evaluate the equilibrium conditions at this benchmark to gain a sense of scale. Substituting $\rho = \frac{1}{2}$ gives $\mu = \frac{1}{2}(N-1)$ and $\sigma = \frac{1}{2}\sqrt{N-1}$. Take the equilibrium relationship, $z = (x - \mu)/\sigma = (\sigma - 1)/x$, and then express it as a quadratic in x: $x(x - \mu) - \sigma(\sigma - 1) = 0$. We can then solve for the positive root of this quadratic, $x > 0$, using the standard formula, $x = \left(\mu + \sqrt{\mu^2 + 4\sigma(\sigma - 1)}/2\right)$, with the discriminant as $\mu^2 + 4\sigma(\sigma - 1)$.

To facilitate the analysis, we will make the following approximation

$$\sqrt{\mu^2 + 4\sigma(\sigma - 1)} = \mu\sqrt{1 + 4\sigma(\sigma - 1)/\mu^2} \simeq \mu\left(1 + 2\sigma(\sigma - 1)/\mu^2\right)$$

Plugging this into the root, we have

$$x \simeq \frac{\mu + \mu\left(1 + 2\sigma\left(\sigma - 1\right)/\mu^2\right)}{2} = \mu + \frac{\sigma\left(\sigma - 1\right)}{\mu}.$$

So the level difference is

$$x - \mu \simeq \frac{\sigma\left(\sigma - 1\right)}{\mu}$$

With $\rho = \frac{1}{2}$, $\sigma\left(\sigma - 1\right)/\mu = \frac{1}{2} - \frac{1}{\sqrt{N-1}}$, and hence

$$x - \mu \simeq \frac{1}{2} - \frac{1}{\sqrt{N-1}}$$

Thus, $x \simeq \mu$ up to a small positive offset of order less than $\frac{1}{2}$, and $z = \left(\sigma - 1\right)/x \simeq 1/\sqrt{N} \to 0$ as N becomes large.

The large-N analysis clarifies the equilibrium structure of the model. When the number of countries is large, the foreign ratification margin becomes vanishingly small, placing the agreement at a balanced interior point where the expected and required ratifications almost coincide. This delivers three insights.

First, equilibrium pivotality is self-stabilizing: both lobby groups remain active because the contest is finely balanced, producing an aggregate ratification probability close to one-half. Second, the government's design problem reduces to maintaining that balance. It adjusts stringency and the MRT to leverage both group's incentive to lobby. Third, because the result relies only on symmetry, diminishing returns to lobbying, and large-N aggregation, it is not sensitive to the precise functional forms of the contest success function or of lobbying technology. The large-N solution therefore provides a transparent illustrations of the tensions captured by the model.

5.2 Participation in the IEA Formation Process

Having characterized equilibrium IEA design, we now examine the participation decision. The key question is whether governments prefer to join the IEA formation process rather than free-ride on others' efforts. To examine this, we now extend the model by adding an initial participation stage, which we will refer to as Stage 0. This stage will allow countries to choose whether or not they want to participate in the IEA, anticipating the way the IEA formation process will play out in the three further stages modeled above.

At Stage 0, all governments are invited to participate in the IEA formation process, and those who do not wish to do so can opt out. This design ensures that inefficient equilibria do not arise as a result of coordination failure. It is a reasonable assumption given that

governments can communicate over the costs and benefits of participation in the IEA before committing to participation.

Governments that decide to participate in the IEA formation process at Stage 0 proceed to the design stage (choosing γ and x in Stage 1), the lobby stage (where l_{Ai} and $l_{P,i}$ are made in Stage 2) and the ratification stage (Stage 3) as specified above; those that do not participate remain permanently at the status quo policy benchmark $e_{SQ} = \beta$, and not lobbying takes place in those countries. We will ask whether full participation, with all N governments entering the IEA formation process, is immune to unilateral deviations in Stage 0, anticipating the subsequent Stages 1-3.

Let s denote the number of governments that participate in the IEA formation process, where $2 \leq s \leq N$. If s governments enter at Stage 0, they choose an optimal pair $(\gamma(s), x(s))$ in Stage 1. At Stage 2, a parallel contest then takes place in each of the s participating countries, determining lobby contributions $l_{A,i}$, $l_{P,i}$, and the corresponding ratification probability $\rho(s)$ in each country of those countries, as above. At Stage 3, ratification votes occur. the agreement enters into force if and only if at least $M(s) = 1 + x(s)$ countries ratify the IER.

If country i is one of the s participates in the IER formation process, its payoff structure is given by (27), as we now show. Begin with the structural form:

$$\mathbb{G}_i^{\text{in}}(s) = \rho(s) v_{P,i}(s) + \Theta_i(s),$$

where $v_{P,i}(s)$ is the value to the pro-IEA group of an IEA with s participants going ahead, and $\Theta_i(s) = [e_{SQ} - e_{IA}(s)] \sum_{r=M(s)}^{s-1} r f(r; s-1, \rho(s))$. Since $e_{SQ} - e_{IA}(s) = \gamma(s)$, this becomes

$$\mathbb{G}_i^{\text{in}}(s) = \rho(s) v_{P,i}(s) + \gamma(s) \sum_{r=M(s)}^{s-1} r f(r; s-1, \rho(s)).$$

The valuation term $\rho(s) v_{P,i}(s)$ captures the benefit to government i when it ratifies and the IEA enters into force, while the second term represents the expected free-riding benefit from the other $s-1$ countries in the IEA reducing their emissions.Since, as shown by (**??**), $\mathbb{G}_i^{\text{in}}(s)$ has the same structural form as $\mathbb{G}_i$, we may apply Proposition 1 to write

$$\mathbb{G}_i^{\text{in}}(s) = \gamma(s) \sum_{r=M(s)}^{s} r f(r; s, \rho(s)).$$

Under the normal approximation, $\sum_{r=M(s)}^{s} r f(r; s, \rho(s)) \simeq \rho(s) s \bar{G}(z(s))$, where $z(s)$ is the standardized ratification margin induced by the equilibrium $(\gamma(s), x(s), \rho(s))$. Hence we

write

$$\mathbb{G}_i^{\text{in}}(s) = \gamma(s)\rho(s)s\bar{G}\left(z(s)\right). \tag{33}$$

The derivation of (33) is exactly as above for (27), except that s replaces N throughout.

The interpretation of $\mathbb{G}^{\text{in}}(s)$ is therefore the same as for $\mathbb{G}$, except that there are now s participants in the process rather than N. Here, $\rho\left(s\right)s$ is the expected number of ratifying countries, while multiplying by $\gamma\left(s\right)\bar{G}(z\left(s\right))$ gives the expected welfare gain from emissions reductions when the IEA between s countries enters into force. Note that the status-quo emissions $e_{SQ} = \beta$ remain fixed both in the status quo and in the IEA, so when welfare is measured relative to the status-quo benchmark their emissions cancel out. For this reason they do not appear in $\mathbb{G}^{\text{in}}(s)$.

To determine the payoff of a non-participant government i, that chooses not to participate in the IEA formation process when s other countries do, we use the structural form in (**??**) together with the normal approximation and Corollary 1. A non-participant never ratifies, and hence does not receive the valuation term $v_{P,i}$, so its payoff is given by the free-riding benefit alone:

$$\mathbb{G}^{\text{out}}(s) = \Theta_i(s).$$

Applying Corollary 1, for any non-participant i facing an IEA of size s,

$$\Theta_i(s) = \gamma(s)\sum_{r=M(s)}^{s} rf(r;s,\rho(s)) - \rho(s)v_{P,i}\left(s\right). \tag{34}$$

Using the normal approximation,

$$\mathbb{G}^{\text{out}}(s) = \rho(s)s\bar{G}\left(z(s)\right) - \rho(s)v_{P,i}\left(s\right). \tag{35}$$

Then by (33) and (35),

$$\mathbb{G}^{\text{in}}(s) - \mathbb{G}^{\text{out}}(s) = \rho(s)v_{P,i}\left(s\right),$$

which reflects the fact that only participant governments obtain $\rho(s)v_{P,i}\left(s\right)$; non-participant governments free-ride on emissions reductions but do not receive the payoff from lobbying.

In comparing the payoff to government i of participation in the IEA versus non-participation, it is important to note that they are comparing $\mathbb{G}^{\text{in}}(s)$ to $\mathbb{G}^{\text{out}}(s-1)$, since the size of the IEA is smaller by one if country i does not participate. To establish that $\mathbb{G}^{\text{in}}(s) > \mathbb{G}^{\text{out}}(s-1)$, it is sufficient to show that $\Theta_i(s)$ is increasing in s, since we already know that $\rho(s)v_{P,i}\left(s\right) > 0$.

In order to characterize and compare $\mathbb{G}^{\text{in}}(s)$ to $\mathbb{G}^{\text{out}}(s-1)$, we will first characterize equilibrium for an IEA with s members. As we now explain, the equilibrium has the same characterization as for the large-N case presented in Proposition 4, providing that s, like N,

is large.

The reason the equilibrium conditions for an s–country IEA mirror those of the full N-country case is that government i's optimization problem depends only on two objects: (i) the expected number of other $s-1$ members, and (ii) the variability of that number. In the N-country case, these are determined by a binomial distribution with $N-1$ trials; for an IEA of size s, they are determined by a binomial distribution with $s-1$ trials. Once a government takes the IEA size s as given, nothing else in the structure of the design problem changes: the CSF has the same form, the government's payoff depends on (γ, x) only through the mean and dispersion of foreign IEA member variables, and the equilibrium condition that trades off stringency against pass probability is unaffected by how many countries remain outside. Thus replacing $N-1$ with $s-1$ simply rescales the mean and variance of the foreign ratification distribution, but leaves the logic of the equilibrium intact. The same three forces, namely expected participation, dispersion, and endogenous pivotality, therefore generate exactly the same system of equations that characterize $(\gamma(s), x(s), \rho(s), z(s))$ for any size of IEA s. We present this argument formally as follows.

Proposition 5. *Fix $s \in \{2, \ldots, N\}$ and suppose that at Stage 0 exactly s governments enter the IEA formation process. Conditional on s, the Stage 1-3 subgame among these s countries has a symmetric interior equilibrium $(\gamma(s), x(s), \rho(s))$ whose structure is identical to the N–country case. In particular:*

1. *The equilibrium condition is*

$$z(s) = \frac{x(s) - \mu_s(\rho(s))}{\sigma(\rho(s))} = \frac{\sigma(\rho(s)) - 1}{x(s)}.$$

2. *The contest success function condition is*

$$\rho(s) = \frac{R\big(x(s), z(s)\big)}{R\big(x(s), z(s)\big) + \gamma(s)}.$$

3.

$$\rho = \frac{1}{2}\left[1 - \frac{\sqrt{2\pi\sigma^2}}{(s-1)}\right] \to \frac{1}{2} \text{ as } s \text{ becomes large.}$$

$$x = \frac{1}{2}\left[(s-1) - \sqrt{2\pi\sigma^2}\right]$$

$$\gamma = \sqrt{\frac{2}{\pi}}\left[\frac{\sigma}{\rho^2}\right]$$

In short, the form of the equilibrium is universal; only the number of potential foreign ratifiers changes from $N-1$ to $s-1$, and the entire equilibrium structure adjusts mechanically through this substitution.

Having characterized equilibrium, we now need to evaluate how equilibrium values change with s. Despite the fact that the equilibrium solutions presented in Proposition 6 are not closed form, we can nevertheless obtain comparative statics that show how the equilibrium values respond to a change in s.

Proposition 6. *Assume an equilibrium as characterized in Proposition 5. The equilibrium values respond to a change in s as follows:*

$$\frac{(s-1)}{\rho}\frac{d\rho}{ds} = (1-2\rho)(1-\rho) > 0$$
$$\frac{(s-1)}{x}\frac{dx}{ds} = [1+(1-2\rho)(1-\rho)] > 0$$
$$\frac{(s-1)}{\gamma}\frac{d\gamma}{ds} = \left[2\rho^2-(1-2\rho)^2\right] > 0$$

Combining the characterization of ρ presented in Propositions 5 and 6, we have that in equilibrium, $\rho \in \left(\frac{1}{3}, \frac{1}{2}\right)$ and it tends towards $\frac{1}{2}$ as s becomes large. The fact that ρ lies in this range in equilibrium determines the positive relationship between γ and s.

We are now ready to show that $\Theta_i(s)$ is increasing in s. By (34) and (35),

$$\Theta_i(s) = \gamma(s)\rho(s)s\overline{G}\big(z(s)\big) - \rho(s)v_{P,i}. \tag{36}$$

By Propositions 6 and 7, as s becomes large, in equilibrium we have

$$\rho(s) \to \tfrac{1}{2}, \qquad z(s) \to 0, \qquad \gamma(s) \to \gamma^* > 0,$$

and hence

$$\overline{G}\big(z(s)\big) \to \overline{G}(0) = \tfrac{1}{2}.$$

Now consider the discrete difference

$$\Theta_i(s) - \Theta_i(s-1).$$

Using the large-s approximation developed above, recall that $z(s) \to 0$ and therefore $G(z(s)) \to G(0) > 0$ as s becomes large. It follows that for sufficiently large s the discrete difference $\Theta_i(s) - \Theta_i(s-1)$ is arbitrarily close to the positive constant $G(0)\gamma^*/2$. Hence, there exists a value $\bar{s}$ such that $\Theta_i(s) - \Theta_i(s-1) > 0$ for all $s \geq \bar{s}$. This establishes that $\Theta_i(s)$ is strictly

increasing for all sufficiently large s.

Proposition 7. *Assume the normal approximation applies and the large–s equilibrium described in Proposition 6 obtains, so that along the equilibrium locus we have*

$$\rho(s) \to \tfrac{1}{2}, \qquad z(s) \to 0, \qquad \gamma(s) \to \gamma^* > 0 \quad \textit{for large } s.$$

Then there exists $\bar{s} \geq 2$ such that for all $s \geq \bar{s}$ the free–riding benefit $\Theta_i(s)$ is strictly increasing in s:

$$\Theta_i(s) > \Theta_i(s-1).$$

This result is interesting in its own right. It confirms that the benefit to each country from free-riding on IEA formation is increasing in the number of countries in the IEA, as one would expect. But, taken together with the fact that $\mathbb{G}^{\text{in}}(s) - \mathbb{G}^{\text{out}}(s) = \Theta_i(s)$, it also confirms that $\mathbb{G}^{\text{in}}(s) > \mathbb{G}^{\text{out}}(s-1)$ as required to establish that any government i would prefer to be part of an IEA of size s than to remain outside of it. Since this holds for $N = s$, we can now characterize the stability in equilibrium of full participation.

Proposition 8. *There exists a unique Subgame Perfect Nash Equilibrium of the game with Stages 0-3, in which all N countries participate in the IEA formation process.*

The uniqueness of equilibrium depends on our assumption that at Stage 0, the process starts with all countries assumed to be involved and then given the opportunity to opt out. However, we think this assumption is reasonable because it removes the possibility of multiple equilibria through coordination failure. To put this another way, we assume that the institutional arrangements put in place to form an IEA give countries the opportunity to communicate over the formation of an IEA. Then failure to participate would only arise if this were individually rational. This establishes that full participation is not only stable, but uniquely stable, making the IEA self-enforcing despite free-riding benefits and lobbying distortions.

5.3 Probability of *Something* passing

Suppose the objective is to choose γ and x in order to increase the probability of an "agreement" passes. In other words

$$\Upsilon = \bar{F}(M-1, N, \rho) \tag{37}$$

Differentiating

$$\frac{\partial \Upsilon}{\partial \gamma} = \frac{M}{\rho} f(M, N, \rho) \frac{\partial \rho}{\partial \gamma} < 0$$

$$\frac{\partial \Upsilon}{\partial M} = f(M, N, \rho)\left[\frac{M}{\rho}\frac{\partial \rho}{\partial M} - 1\right]$$

THIS last condition doesn't really matter, because the agreement will have $\gamma = 0$ and thus will be no real agreement.

6 The Effect of Lobby Revenue

A more general representation of the government's objective function is

$$\begin{aligned}\mathbb{G} &= \left[\underbrace{\rho v_A - \mathbb{E}h}_{l} + a\rho(v_P - v_A) + a\Theta\right] \\ &= a\left[\frac{(1-a)}{a}\rho v_A - ab\left|v_P - v_A\right| + \Phi\right]\end{aligned}$$

What does this mean??? First assume that participation constraints don't bind with head-starts. Lobby revenue helps offset domestic damages from reducing emissions and exactly offsets when $a = 1$. Having head-starts introduces lobby-leakages – Note that introducing head-starts does not affect ratification uncertainty as it continues to be the case that $\rho = \frac{v_P}{v_P + v_A}$. This all means that adding all of these things only changes what the government would choose for the agreement, not the underlying ratification uncertainty and thus not the agreement chosen that maximizes expected emission reduction.

Proposition 9. *More weight on expected social welfare and less on lobby revenue moves the agreement away from the one that maximizes expected emission reduction.*

Proof. Suppose there was no lobby revenue – then the government's objective function

$$\mathbb{G} = a\left[\Phi - \rho v_A)\right]$$

The FOC for γ is

$$\frac{\gamma}{\mathbb{G}}\frac{d\mathbb{G}}{d\gamma} = 1 - \frac{a\rho v_A}{\mathbb{G}} - \frac{\rho}{\mathbb{G}}\frac{\partial \mathbb{G}}{\partial \rho}\lambda(1-\rho)$$

Evaluating this at the optimum from when $a = 1$ – i.e. $\frac{\rho}{\mathbb{G}}\frac{\partial \mathbb{G}}{\partial \rho}\lambda(1-\rho) = 1$

$$\frac{\gamma}{\mathbb{G}}\frac{d\mathbb{G}}{d\gamma} = -\frac{a\rho v_A}{\mathbb{G}} < 0$$

Essentially for any given threshold M, the objective function is shifted to the left. $\square$

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Appendix

Proof of Lemma 1. We can rewrite equation (20) as

$$\rho = \frac{2\left[1+(M-1)\mathrm{pr}\left(X=M-1|X\geq M-1;\rho\right)\right]}{2\left[1+(M-1)\mathrm{pr}\left(X=M-1|X\geq M-1;\rho\right)\right]+\gamma} \equiv \Omega(\rho). \tag{38}$$

By the intermediate value theorem, there is a unique interior fixed point of $\Omega\left(\rho\right):[0,1]\times[0,1]$ if (i) $\Omega\left(\rho\right)$ is continuously decreasing in ρ and (ii) $0<\lim_{\rho\to 1}\Omega(\rho)<\lim_{\rho\to 0}\Omega(\rho)<1$.

First, consider (ii). All of the mass in the binomial distribution shifts to the highest possible number of successes as the binomial success probability ρ approaches 1. Formally,

$$\lim_{\rho\to 1}\mathrm{Pr}(X=M-1|X\geq M-1;\rho)=0 \tag{39}$$

given that

$$\begin{aligned}\lim_{\rho\to 1}\mathrm{Pr}(X=M-1;\rho) &= \lim_{\rho\to 1}\binom{N-1}{M-1}\rho^{M-1}\left(1-\rho\right)^{N-M}\\ &= 0 \qquad (40)\\ \lim_{\rho\to 1}\mathrm{Pr}(X\geq M-1;\rho) &= 1-\lim_{\rho\to 1}\sum_{i=0}^{M-1}\binom{N-1}{i}\rho^{i}\left(1-\rho\right)^{N-1-i}\\ &= 1. \qquad (41)\end{aligned}$$

Similarly, conditional on realizing at least $M-1$ successes, all of the mass in the binomial distribution shifts to the lowest number of possible successes $M-1$ as the binomial success probability approaches 0. Formally,

$$\begin{aligned}\lim_{\rho\to 0}\mathrm{Pr}(X=M-1|X\geq M-1;\rho) &= \lim_{\rho\to 0}\frac{\binom{N-1}{M-1}\left(1-\rho\right)^{N-M}}{\sum_{i=0}^{M-1}\binom{N-1}{i}\rho^{i-(M-1)}\left(1-\rho\right)^{N-1-i}} \qquad (42)\\ &= \frac{\binom{N-1}{M-1}}{\binom{N-1}{M-1}}\\ &= 1. \qquad (43)\end{aligned}$$

Hence, using equation (20),

$$0<\lim_{\rho\to 1}\Omega(\rho)=\frac{2}{2+\gamma}<\lim_{\rho\to 0}\Omega(\rho)=\frac{2M}{2M+\gamma}<1.$$

Second, consider (i). $\Omega\left(\rho\right)$ is continuously increasing in $\mathrm{pr}\left(X=M-1|X\geq M-1;\rho\right)$

using equation (20). And, using equation (42),

$$\Pr(X = M-1 | X \geq M-1; \rho) = \binom{N-1}{M-1} \left[\sum_{i=M-1}^{N-1} \binom{N-1}{i} \left(\frac{\rho}{1-\rho} \right)^{i-(M-1)} \right]^{-1} \tag{44}$$

is continuously decreasing in ρ. Thus, $\Omega(\rho)$ is continuously decreasing in ρ.

The comparative static results follow from implicitly differentiating $\rho - \Omega(\rho) = 0$ after establishing that, for a fixed ρ, $\frac{\partial \Omega(\rho)}{\partial \gamma} < 0$ and $\Omega(\rho)$ is increasing in M. Inspection of equation (38) immediately reveals $\frac{\partial \Omega(\rho)}{\partial \gamma} < 0$. For $\Omega(\rho)$ increasing in M, we need to establish $\mathrm{pr}(X = M-1 | X \geq M-1; \rho)$ is increasing in M. Intuitively, conditional on realizing at least $M-1$ successes, all of the mass in the binomial distribution shifts to the lowest number of possible successes $M-1$ as the threshold number of successes $M-1$ grows. Formally, log-concave probability mass functions are increasing in M (Bagnoli and Bergstrom, 2005, Corrolary 2) and the probability mass functions of the binomial distribution is log-concave (Keilson and Gerber, 1971, p.386).

Given $\rho(\gamma, M)$ is increasing in M on the interval $[1, N]$, we have $\rho(M+1) \geq \rho(M)$ for any integer $M \in \{1, \ldots, N-1\}$. Thus, we have our desired comparative static result for continuous and discrete M.

A General Properties

A.1 Binomial Distribution

Define $x \equiv (M-1)$

$$
\begin{aligned}
f(r, N-1) &= \frac{(N-1)!}{r!(N-1-r)!}\rho^r(1-\rho)^{N-1-r} \\
\bar{F}(x-1, N-1) &= \sum_{r=x}^{N-1} f(r, N-1) \\
\Theta &= \gamma \sum_{r=M}^{N-1} f(r, N-1) r \\
&= \gamma\rho(N-1)\bar{F}(x-1, N-2) \\
f(r, N-2) &= \frac{(N-2)!}{r!(N-2-r)!}\rho^r(1-\rho)^{N-2-r} \\
&= \frac{(N-1-r)}{(1-\rho)(N-1)}\frac{(N-1)!}{r!(N-1-r)!}\rho^r(1-\rho)^{N-1-r} \\
&= \left(\frac{(N-1-r)}{(1-\rho)(N-1)}\right) f(r, N-1) \\
\frac{\partial f(x)}{\partial \rho} &= \left[\frac{x}{\rho} - \frac{(N-1)-x}{(1-\rho)}\right] f(x) \\
\frac{\partial \bar{F}(x-1, N-1)}{\partial \rho} &= (N-1) f(x-1, N-2) \\
&= (N-1)\frac{(N-2)!}{(x-1)!(N-1-x)!}\rho^{x-1}(1-\rho)^{N-1-x} \\
&= \frac{x}{\rho} f(x, N-1) \\
\frac{\partial \bar{F}(x-1, N-2)}{\partial \rho} &= (N-2) f(x-1, N-3) \\
&= \frac{x}{\rho} f(x, N-2) \\
&= \frac{x(N-1-x)}{\rho(1-\rho)(N-1)} f(x, N-1) \\
\frac{\partial \tilde{f}(x)}{\partial \rho} &= \frac{1}{\bar{F}(x-1)^2}\left[\bar{F}(x-1)\frac{\partial f(x)}{\partial \rho} - f(x)\frac{\partial \bar{F}(x-1)}{\partial \rho}\right] \\
&= \frac{\tilde{f}(x)}{\rho}\left[\left[\frac{x-\mu}{(1-\rho)}\right] - x\tilde{f}(x)\right]
\end{aligned}
$$

A.2 Normal Distribution

Define

$$x \equiv (M-1) \qquad z \equiv \frac{x-\mu}{\sigma} \qquad \mu = \rho(N-1)$$
$$\sigma = \sqrt{(1-\rho)\mu} \qquad f(x) = g(z) = \frac{\phi(z)}{\sigma} \qquad \bar{F}(M-2) = 1 - G(z) \equiv \bar{G}(z) = \bar{\Phi}(z)$$

$$\frac{\partial g}{\partial x} = -g(z)\left(\frac{z}{\sigma}\right) \qquad \frac{\partial g(\cdot)}{\partial \mu} = g(z)\left(\frac{z}{\sigma}\right) \qquad \frac{\partial g(\cdot)}{\partial \sigma} = g(z)\left(\frac{z^2-1}{\sigma}\right)$$
$$\frac{\partial \bar{G}}{\partial x} = -\frac{\partial G}{\partial x} = -g(z) \qquad \frac{\partial \bar{G}}{\partial \mu} = g(z) \qquad \frac{\partial \bar{G}}{\partial \sigma} = zg(z)$$
$$\frac{\partial \sigma}{\partial \rho} = \frac{\mu}{\sigma}\left(\frac{(1-2\rho)}{2\rho}\right) \qquad \frac{\partial \sigma^2}{\partial \rho} = \frac{\mu(1-2\rho)}{\rho} \qquad \frac{\partial \mu}{\partial \rho} = \frac{\mu}{\rho}$$

So

$$\frac{\rho}{\bar{G}(z)}\frac{\partial \bar{G}(z)}{\partial \rho} = \mu \tilde{g}(z)\left[1 + \frac{z}{\sigma}\left(\frac{(1-2\rho)}{2}\right)\right]$$
$$\frac{\rho}{g(z)}\frac{\partial g(z)}{\partial \rho} = \mu\left[\frac{z}{\sigma} + \left(\frac{z^2-1}{\sigma^2}\right)\left(\frac{(1-2\rho)}{2}\right)\right]$$

B Extra